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Towards Sustainable Farming: Leveraging AIoT for Precision Water Management and Crop Yield Optimization

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Abstract

Sustainable Development Goals (SDGs) by the United Nations have Zero Hunger and Responsible Consumption and Production as the two significant agriculture-related components. Conventional irrigation control systems, relying heavily on manual labour, exhibit shortcomings in minimizing farmer input dependency. Recognizing the critical interdependence of water resource management, crop yield optimization, and environmental preservation, this work contends that precision in water management is essential. While efforts to reuse treated wastewater offer potential water conservation solutions, they often compromise crop yield. Precisely calculating water requirements can enhance crop yield, reduce human involvement, eliminate potential decision-making errors, and address water conservation issues. Automating tasks like irrigation demands comprehensive knowledge of crop and soil characteristics, climate conditions, time factors, and physical parameters like temperature, soil moisture, and humidity. This research addresses the limitations of current agricultural practices by advocating for a paradigm shift towards sustainable farming through the integration of Artificial Intelligence of Things (AIoT). The proposed framework leverages the fusion of the Artificial Intelligence (AI) and Internet of Things (IoT) to automate irrigation by employing a comprehensive analysis of diverse parameters. Through empirical evaluation, this work demonstrates that AIoT-based precision water management not only improves crop yield but also reduces human intervention, addressing water conservation challenges, and fostering sustainable agriculture practices.

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Keywords: Artificial Intelligence, Artificial Intelligence of Things (AIoT), Internet of Things (IoT), Precision Water Management, Sustainable Development Goals (SDGs)

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1. Introduction

Throughout human history, agriculture has stood as a fundamental pillar of civilization, providing the necessary sustenance and resources crucial for human survival. However, as our global population continues to expand and the ramifications of climate change intensify, the demand for innovative solutions to enhance food production, minimize waste, and optimize resource utilization has become increasingly pressing. The Sustainable Development Goals (SDGs) [1] by the United Nations focuses on Zero Hunger (SDG 2) and Responsible Consumption and Production (SDG 12). In this scenario, the realm of Artificial Intelligence (AI), a discipline within computer science focused on creating intelligent machines capable of human-like tasks, emerges as a promising frontier for revolutionizing agricultural practices. In the agricultural context, AI demonstrates its potential by harnessing extensive datasets encompassing soil quality, weather patterns, and crop yields, offering unprecedented insights and tools that were once beyond reach. This potential holds the key for enhancing efficiency, minimizing waste, and refining resource management within the agricultural sector [2]. At present, AI finds application in various facets of agriculture, including but not limited to crop monitoring, precision agriculture, livestock management, and optimization of supply chains. The applicability of AI in agriculture is expansive, holding the promise of significant advancements across diverse domains in the years to come. Crucially, the incorporation of AI into agriculture is poised not only to elevate food production and mitigate waste, but also to streamline resource utilization.

On the other hand, Internet of Things (IoT) has evolved as a transformative technology, enabling extensive research and practical applications across diverse domains. Its seamless connectivity and data exchange capabilities have spurred innovation, driving advancements in efficiency and resource optimization. In agriculture, IoT optimizes resource management by efficiently harnessing data from sensors, providing farmers with real-time insights to make informed decisions on irrigation, fertilization, and crop health [3].

The integration of the AI and IoT, also known as Artificial Intelligence of Things (IoT), into real-time processes further amplifies the potential for data-driven decision-making and real-time monitoring. The synergy between AI and IoT is particularly useful in the efficient management of agricultural ecosystems, ensuring a more sustainable and efficient agricultural landscape. The infusion of AIoT not only reduces human effort but also mitigates the risk of potential losses resulting from errors introduced by human factors in critical decision-making areas [4]. AIoT systems streamline data acquisition by deploying IoT devices with various sensors that collect real-time information. This data is transmitted to cloud platforms, where AI algorithms analyze it for insights, facilitating predictive analytics and anomaly detection. Automation in decision-making is achieved through rule-based systems, while continuous learning and improvement are driven by feedback loops, enhancing the system's adaptability over time.

The prevailing farming systems in use today exhibit notable deficiencies, such as lack of self-sufficiency, reliance on limited data from studies, and disparity from practical needs. Sustainable agricultural practices in water management involve techniques like drip irrigation, rainwater harvesting, and efficient water use through precision farming. Precision farming employs technology to optimize resource utilization, minimizing water wastage and maximizing crop yields. These practices are crucial for conserving water resources, mitigating the impact of drought, and ensuring the long-term viability of agriculture in the face of water scarcity and climate change. Accurate data and a true representation of the farming situation are essential for sound decision-making in Sustainable agricultural practices. Decisions crucial to farming success, such as determining the appropriate amount and timing of irrigation, assessing soil moisture levels, and other pivotal factors, are susceptible to errors. IoT presents a solution by enabling real-time data monitoring, providing farmers with timely information. Meanwhile, AI has the potential to aid in making accurate decisions based on the information received through IoT devices, such as sensor arrays strategically placed across farm fields [5], [6]. Thus, AIoT offers a transformative pathway towards precision agriculture, minimizing uncertainties, and optimizing outcomes for farmers.

To address these challenges, the proposed work aims to develop an AIoT system that applies AI algorithms to predict the water requirement for different crops based on soil, weather, and crop information captured in real-time using IoT. The goal is to optimize crop yield, as well as automate the agricultural practices. The work is directed towards advancing sustainable agriculture by supporting water conservation, automation, efficiency, and informed decision-making for farmers. Following are the key contributions of this work:

1. This research contributes by introducing a water release prediction system designed for farmers that minimizes water wastage, ensuring the optimal utilization of water resources in agriculture.
2. The proposed AIoT system automates data acquisition and analysis, thereby eliminating manual intervention and reducing the workload on farmers.
3. The AI-based approach presented in this work empowers farmers to make informed decisions about irrigation, thus addressing Sustainable Development Goals 2 and 12.

This paper is organized as follows: Section 2 summarizes the existing work related to the topic. Section 3 outlines the proposed AIoT-driven approach to predict crop water requirements. Empirical validation of the proposed approach is stated in Section 4. Section 5 concludes the work and presents directions for future research.

2. Related Work

The integration of Artificial Intelligence (AI) into agriculture has a rich history dating back to the 1980s, when researchers and farmers embarked on exploring its potential applications in crop management and precision farming. One of the earliest uses of AI in agriculture was for image recognition, specifically in identifying plant diseases and pests. Researchers developed algorithms that could analyze images of crops and accurately identify the type of disease or pest present. This empowered farmers to swiftly detect and address crop issues, leading to increased yields and reduced losses. As computing power increased and Machine Learning algorithms became more sophisticated, the use of AI in other areas of agriculture began as well. For example, AI is now used to optimize irrigation schedules, predict weather patterns, and even control autonomous farm equipment. One of the most significant benefits of AI in agriculture is its ability to help farmers make data-driven decisions. By collecting and analyzing data on soil quality, weather patterns, and crop yields, AI provides insights that help farmers optimize their operations and increase productivity. The transformative impact of AI-powered technologies on agriculture is evident in the current landscape. AI-powered technologies have revolutionized agriculture, ushering in a new era of efficiency, sustainability, and productivity.

In literature, precision farming has been attempted, using AI-powered drones and sensors to meticulous monitor crops from planting to harvest, optimizing irrigation, fertilization, and pest control [7]. Researchers [8] have examined the trends in worldwide research on sustainable irrigation in agriculture in the last two decades. They explore the key influencers driving this research and highlight the most prominent topics that have garnered attention during this period.

In the past, researchers have delved into the criteria for irrigation water quality concerning the reuse of wastewater [9]. The study involved a comprehensive review of standards and guidelines from different countries and organizations to propose initial standards for the indirect reuse of wastewater in South Korea.

By analyzing historical weather patterns, soil moisture levels, and other environmental factors, AI algorithms can predict when crops will need water. This allows farmers to adjust irrigation schedules to avoid over- or under-watering, which can lead to reduced crop yields or even crop failure. Researchers [10] have focused on simulating emitter outflow discharge in drip irrigation systems using artificial intelligence models (ANN, NF-SC, NF-FCM, LS-SVM) across varying temperatures and operating pressures. The study finds that all models, particularly LS-SVM, demonstrate acceptable accuracy in estimating the modified discharge coefficient, with the LS-SVM model exhibiting the lowest error.

Integration of Information and Communication Technology (ICT) and IoT have created many more opportunities for multi-format information gathering, processing, storage, retrieval, management, and sharing [11]. Further, the authors in [12] comprehensively review the diverse applications of Artificial Intelligence in agriculture, emphasizing the utilization of sensors and robotics, including drones, for tasks such as irrigation, weeding, and spraying. The examined technologies not only enhance resource efficiency by curbing excess water and pesticide use but also contribute to soil fertility maintenance, efficient manpower utilization, and overall productivity improvement in agriculture.

Another research in this domain [13] introduces a novel framework to enhance the accuracy of crop yield prediction by addressing spatial variability in vegetation and thermal condition indices (VCI and TCI) through higher-order spatial independent component analysis (sICA). Examined in the context of Vietnam, the framework outperforms traditional one-size-fits-all models, demonstrating significant improvements in subregional rice yield forecasts. Furthermore, the proposed PCA-ML combination surpasses ML-only models, offering reliable predictions 1 to 2 months ahead of harvest with minimal error, showcasing its effectiveness.

The researchers in [14] analyze the critical influence of farmer behavior and the impact of both precise and imprecise variables on irrigation depth, proposing a hybrid methodology integrating artificial neural networks, fuzzy logic, and genetic algorithms to model and forecast daily irrigation depths in a real irrigation district in southwest Spain.

The literature [15] also includes comprehensive survey on existing smart farming models for precision agriculture, and reviews innovations in predicting key agrarian practices, highlighting the importance of proper dataset usage, pre-processing, and feature extraction to prevent limitations in smart farming models, ultimately showcasing how AI can automate agricultural practices, enhance crop quantity and quality, and outlines challenges and prospects for future research. Integration of IoT with AI has been proposed as a future work in many of the works.

The subsequent sections discuss the proposed approach and evaluate its performance and effectiveness.

3. Proposed Work

We propose an AIoT system for precision water management and crop yield optimization, as depicted in Fig. 1. It encompasses an infrastructure founded on the Internet of Things (IoT). This infrastructure is designed to mobilize data related to physical parameters, which will then be input into an AI algorithm to predict the water requirements for optimized crop yield.

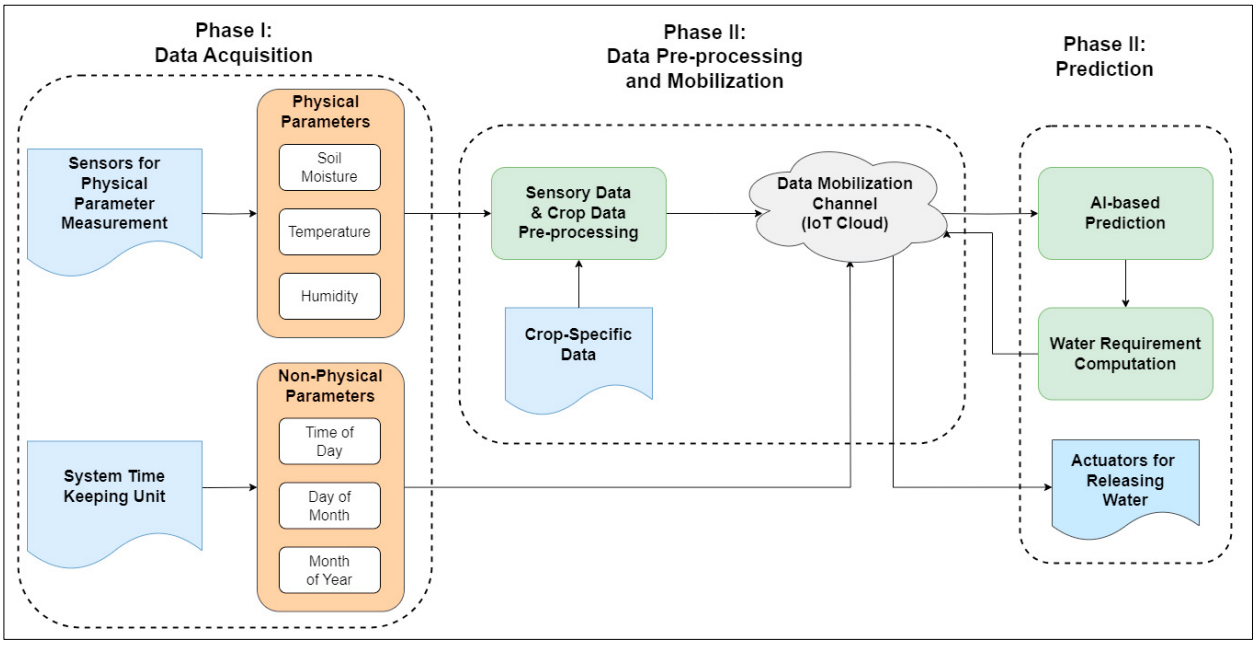


Fig. 1: Proposed Approach for Precision Water Management and Crop Yield Optimization

The system functions in three phases: Data Acquisition, Data Pre-processing and Mobilization, and Prediction. The phase-wise workflow is as follows:

3.1. Phase 1 – Data Acquisition:

The initial stage of the proposed approach entails the acquisition and aggregation of data from diverse physical and non-physical parameters associated with the agricultural setting. These parameters include soil moisture levels, temperature, humidity, sunlight intensity, and other pertinent factors. Deploying sensors and IoT devices in the field enables the continuous collection of this data. Through the constant monitoring of these parameters, the system can acquire real-time information about the agricultural environment, facilitating adaptive and responsive water management strategies.

3.2. Phase 2 – Data Pre-Processing and Mobilization:

The data collected from the various sensors in Phase 1 undergoes preprocessing in Phase 2 to ensure its quality and readiness for analysis. Data preprocessing involves several essential steps, such as data cleaning to remove any outliers or errors, handling missing values through interpolation or imputation techniques, and normalizing the data to bring all parameters to a standardized scale. By preparing the data in this manner, the system ensures that the AI algorithm receives accurate and consistent inputs, thereby enhancing the reliability of the predictions. After preprocessing, the data is transmitted to an IoT cloud platform through a communication channel. The IoT cloud acts as a centralized repository for the data, making it easily accessible for further analysis and modeling. This cloud-based architecture enables seamless integration of data from various sources and allows for the remote access and management of the system.

Phase 3 – Prediction:

In this third and final phase, the pre-processed data from the IoT cloud is utilized by the AI algorithm to analyze the relationships between the various parameters and the water requirement for crops. By considering factors such as soil moisture levels, weather conditions, crop type, and growth stage, the AI algorithm computes the optimal amount of water required for irrigation. The AI algorithm's predictions of water requirements are then fed back to the IoT infrastructure, completing the closed-loop system. In the final phase, the actuators integrated into the irrigation system are programmed to release only the precise amount of water determined by the AI algorithm. This actuator control ensures that each crop receives the appropriate irrigation, minimizing water wastage and supporting efficient water resource management.

By incorporating new data and fine-tuning the AI algorithm, the system continuously learns and adapts to changing environmental conditions, leading to more precise water requirement predictions over time. Effectiveness of the proposed approach for precision water management and crop yield prediction is validated in the next section.

4. Implementation

This section presents the execution of the proposed approach for crop yield optimization, which is subsequently organized into sub-sections covering the IoT elements, data information, the implementation details of AI algorithms, and subsequent performance evaluation as well as water requirement prediction.

4.1. IoT Infrastructure

For implementation of Phase 1 of the proposed approach, the IoT components utilized include Raspberry Pi 3B+, Soil Moisture sensor, Temperature sensor, Solenoid valve, Water Flow Meter, etc. and the connectivity components. Raspberry Pi 3B+ serves as the central processing unit, interfacing with essential components such as Soil Moisture, Temperature sensors, etc. The Solenoid valve regulates water supply, while the Water Flow Meter quantifies usage. Seamless connectivity components ensure real-time data transmission and synchronization, forming a robust foundation for the AIoT-driven precision water management system.

4.2. Training and Real-time Sensory Data

For training the AI algorithms, we utilize the agricultural data accessible from Open Government Data Platform India [16]. The dataset consisted of nearly 50000 records with attributes including Crop Name, Crop Type, Soil Type, Soil Moisture, Temperature, Humidity, Timeline, Rainfall, Area and the amount of required water. Data for

similar attributes is also captured in real-time using the IoT sensors. A sample dataset for Sugarcane crop is presented in Table 1 for clarity.

Table 1: Sample Dataset

Crop	Crop Type	Soil Type	Soil Moisture	Temperature	Humidity	Time of Day	Day of Month	Month of Year	Rainfall (ml)	Area	Water Required (Litres)
Sugarcane	Kharif	Coarse	60	18.8	20	0	1	1	0	1	0
Sugarcane	Kharif	Coarse	60	19.6833	23.1667	1	1	1	0	1	976.2116
Sugarcane	Kharif	Coarse	60	20.5667	26.3333	2	1	1	0	1	1952.4232
Sugarcane	Kharif	Coarse	60	21.45	29.5	3	1	1	0	1	2928.6348
Sugarcane	Kharif	Coarse	60	22.3333	32.6667	4	1	1	0	1	3904.8434
Sugarcane	Kharif	Coarse	60	23.2167	35.8333	5	1	1	0	1	4881.0579
Sugarcane	Kharif	Coarse	60	24.1	39	6	1	1	0	1	5857.2695
Sugarcane	Kharif	Coarse	60	24.9833	42.1667	7	1	1	0	1	6833.4811
Sugarcane	Kharif	Coarse	60	25.8667	45.3333	8	1	1	0	1	7809.6927
Sugarcane	Kharif	Coarse	60	26.75	48.5	9	1	1	0	1	8785.9043
Sugarcane	Kharif	Coarse	60	27.6333	51.6667	10	1	1	0	1	9762.1158
Sugarcane	Kharif	Coarse	60	28.5167	54.8333	11	1	1	0	1	10738.3274
Sugarcane	Kharif	Coarse	60	29.4	58	12	1	1	0	1	11714.5390
Sugarcane	Kharif	Coarse	60	28.5167	54.8333	13	1	1	0	1	10738.3274
Sugarcane	Kharif	Coarse	60	27.6333	51.6667	14	1	1	0	1	9762.1158

4.3. AI Algorithmic and Evaluation Parameters

In the assessment of the most optimal AI algorithm for prediction, a comprehensive approach is adopted. Various regression algorithms that have shown promising results in the literature, including the Decision Tree Regressor, Random Forest Regressor, Support Vector Regressor, Gradient Boost Regressor, and Neural Network Regressor are implemented. These algorithms underwent training using a designated training dataset as mentioned above, and their performance was rigorously evaluated in real-time, utilizing the available data, as well as the data obtained from IoT sensors. Table 2 below describes the hyperparameters used for each of these regression algorithms.

Table 2: Hyperparameters of AI Algorithms

AI Algorithm	Hyperparameters	Values
Support Vector Regressor	Type of Kernel	rbf
	Number of Support Vectors	628
	Use of Heuristic Shrinking Function	False
	Penalty Rate	0.08
	Constant in Decision Function	6.242

	Kernal Coefficient	Scale
Neural Network Regressor	Number of Layers	8
	Type of Layers	Dense, Dropout
	Activation Function	ReLu, Linear
Gradient Boost Regressor	Number of Boosting Stage	300
	Learning Rate	0.04
	Loss Function	Quantile
	Maximum Depth	10
	Quality Measure Function	Friedman MSE
Decision Tree Regressor	Splitter Strategy	Best
	Maximum Depth	10
	Quality Measure Function	Friedman MSE
Random Forest Regressor	Minimal Cost-Complexity Pruning	2.67
	Number of Trees	100
	Quality Measure Function	Friedman MSE
	Randomness of Bootstrapping Method	42

For the prediction (regression) algorithms used in this work, the water requirement is the dependent variable, whereas all others are independent variables. The work utilizes holdout method of evaluation, using two performance evaluation metrics, R-squared and Mean Absolute Percentage Error.

- R-Squared: R-squared [17] is used to evaluate the goodness of fit of a regression model and is mathematically represented as shown in Eq. (1):

$$R^2 = 1 - \frac{SSR}{SST} \quad \ldots.(1)$$

where,
SSR, Sum of Squares of the Residual – represents the sum of the squared differences between the actual observed values of the dependent variable and the predicted values obtained from the regression model;
SST, Total Sum of Squares – signifies the sum of the squared differences between the actual observed values of the dependent variable and the mean of the dependent variable.

- Mean Absolute Percentage Error (MAPE): The MAPE [18] is the average percentage deviation between forecasted and actual values. The absolute deviation is normalized into a percentage, allowing comparison across different data scales. The percentage nature of MAPE makes it easy to interpret. Monitoring MAPE over time helps in identifying trends in forecasting accuracy. If MAPE increases, it may signal the need for model recalibration or improvement. MAPE is mathematically represented as shown in Eq. (2):

$$MAPE = \frac{1}{n}\sum\left|\frac{A-F}{A}\right| 100 \quad \ldots.(2)$$

where,
MAPE is the Mean Absolute Percentage Error;
n is the number of observations or data points;
A is the actual value;
F is the forecasted value.

4.4. Results and Discussion

Fig. 2 and Fig. 3 depict the performance comparison of the considered AI algorithms in predicting the water requirements.

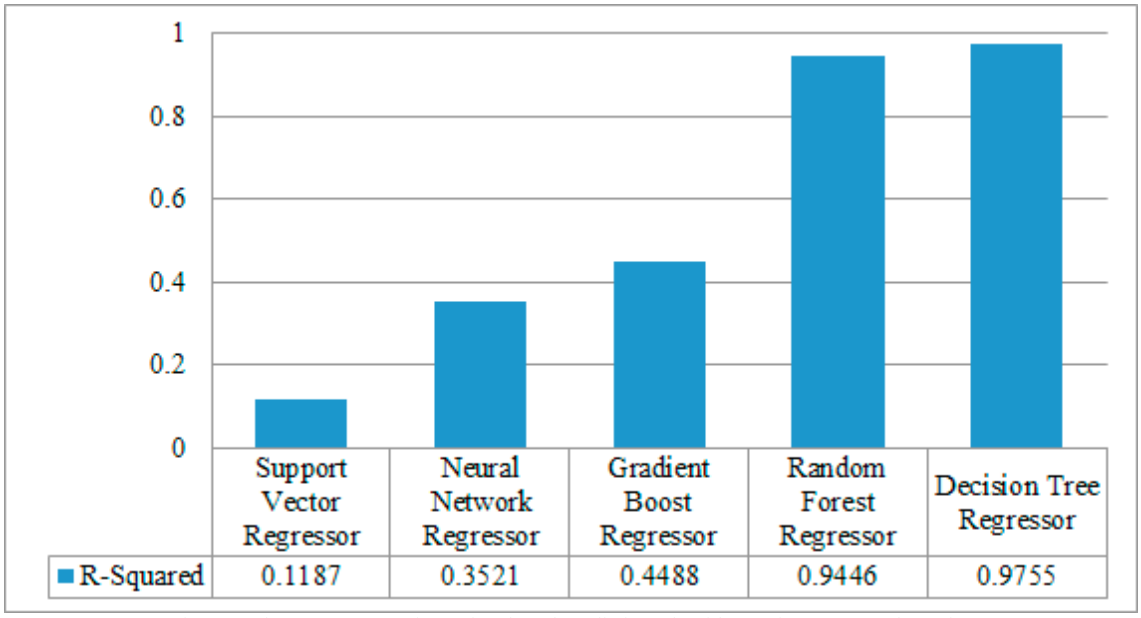


Fig. 2: Performance Comparison of AI-based Prediction Algorithms using R-Squared Metrics

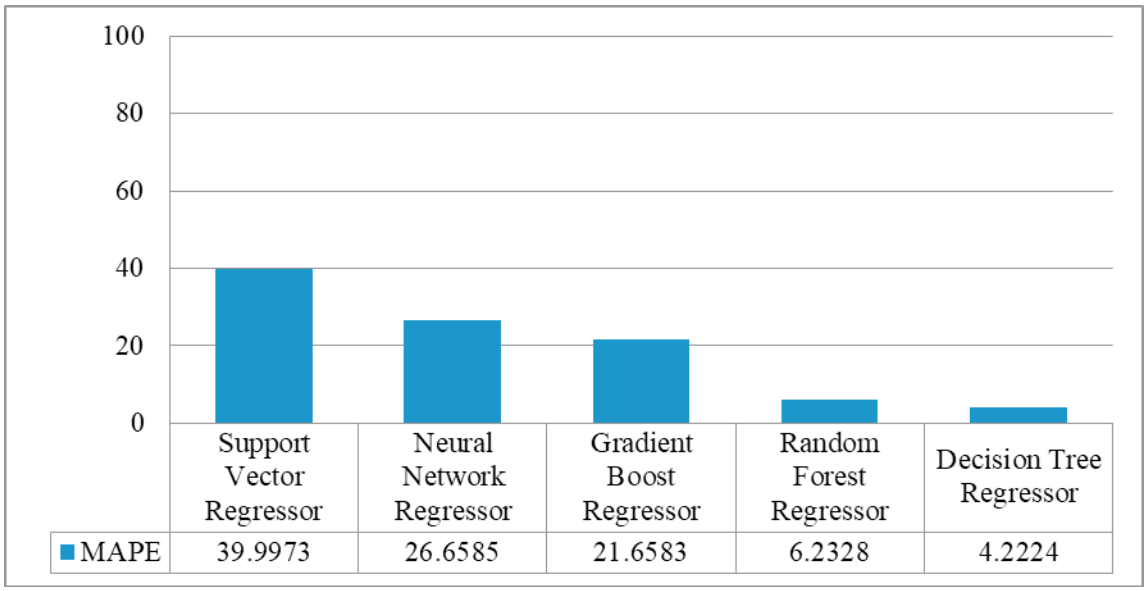


Fig. 3: Performance Comparison of AI-based Prediction Algorithms using MAPE Metrics

As depicted in Figures 2 and 3, the Decision Tree Regressor stands out as the most efficacious model in our evaluation, excelling both in terms of model fitting, as indicated by a superior R-squared value, and in achieving minimal error, as demonstrated by a low Mean Absolute Percentage Error (MAPE). These findings underscore the robust performance of the Decision Tree Regressor, emphasizing its proficiency in accurately capturing the underlying patterns in the data while minimizing predictive discrepancies.

The visual representation of results, followed by the explainability of Decision Tree, as depicted in Fig. 4 demonstrate the potential suitability of Decision Tree Regressor for the predictive tasks under consideration. This sectional visualization of the decision tree model, with its extensive set of rules and conditions, offers valuable insights into how the prediction is made.

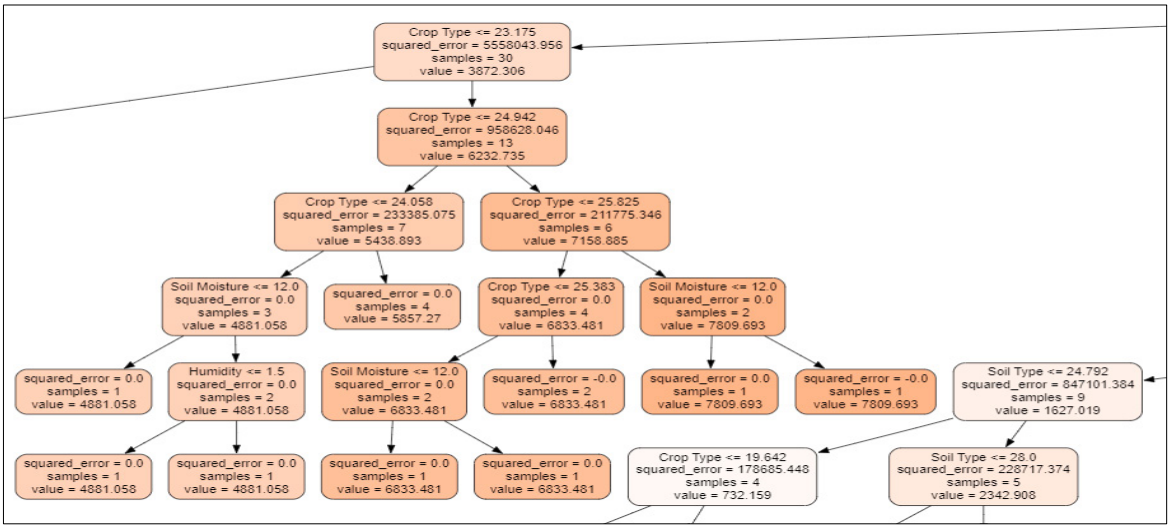


Fig 4: Sectional Visualization of the Decision Tree AI Model

Demonstration of the real-time system that captures data captured from IoT sensors and performs the prediction of water requirement is presented in Fig. 5 and Fig. 6 respectively.

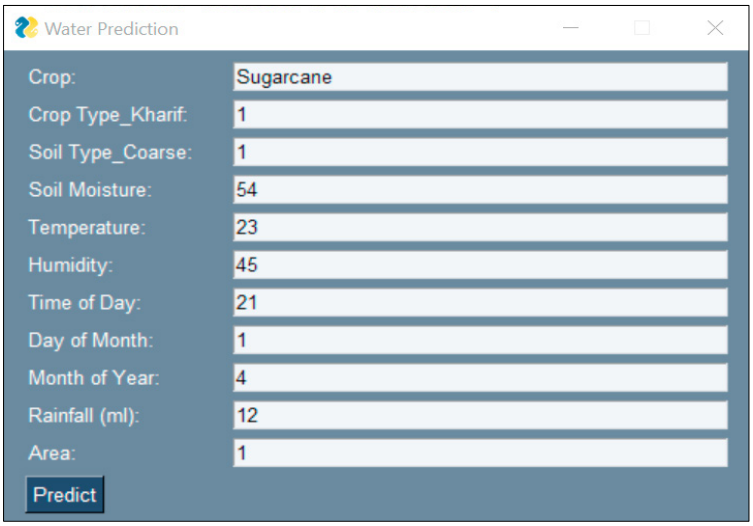


Fig. 5: Data Acquisition from IoT System and Processing

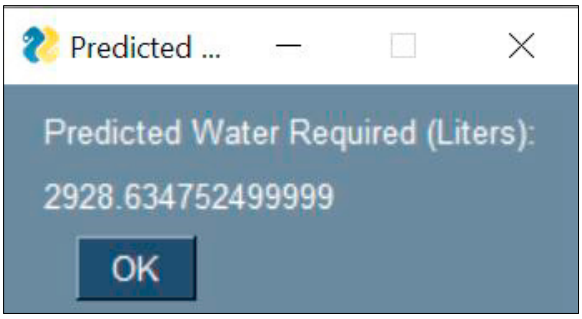


Fig. 6: AI-based Prediction of Water Requirement

5. Conclusion and Future Scope

In conclusion, the proposed AIoT system represents a pivotal step towards sustainable agriculture. By leveraging AI algorithms to predict crop water requirements in real-time through IoT, the work attempts to optimize crop yield, automate agricultural practices, and ensure optimal water resource utilization. The introduced water release prediction system enhances efficiency, reduces manual workload, and empowers farmers with informed irrigation decisions. Empirical evaluation to identify the most suitable AI algorithm as well as real-time incorporation of IoT sensors optimizes irrigation schedules. The seamless integration of AI and IoT not only revolutionizes data acquisition and analysis but also fosters a shift towards precision agriculture.

The future scope of this research is promising, with several avenues for further development and enhancement. Integration with precision agriculture technologies, including remote sensing, satellite imagery, and drones, offers opportunities for more detailed insights into crop conditions. Moreover, establishing platforms or networks for data sharing and collaboration among farmers, researchers, and agricultural experts fosters collective learning and the development of more robust AI models, resulting in improved accuracy, efficiency, and sustainability of irrigation management in agriculture. Overall, integrating cutting-edge technologies and tailoring solutions to specific crops advances the accuracy, efficiency, and sustainability of agriculture.

References

[1] United Nations, Sustainable Development Goals. [Online]. Available: https://www.un.org/sustainabledevelopment. Accessed on March 03, 2023.

[2] Bhat, S., and Huang, N. "Big Data and AI Revolution in Precision Agriculture: Survey and Challenges." IEEE Access 9 (2021). https://doi.org/10.1109/ACCESS.2021.3102227.

[3] Farooq, M., Riaz, S., Abid, A., Umer, T., and Zikria, Y. "Role of IoT Technology in Agriculture: A Systematic Literature Review." Electronics 9, no. 2 (2020). https://doi.org/10.3390/electronics9020319.

[4] Kotecha, R. "Demystifying the Role of AIoT for Sustainable WASH Conditions: Analysis and Research Directions." International Journal of Engineering Trends and Technology 17, no. 10 (2023). https://doi.org/10.14445/22315381/IJETT-V71I10P224.

[5] Kamilaris, A., Kartakoullis, A., and Prenafeta-Boldú, F. "Artificial Intelligence in Agriculture: A Comprehensive Review." IEEE Access (2020). https://doi.org/10.1109/ACCESS.2020.3039159.

[6] Kumbhar, S., Gunjal, B., and Mahalle, P. "Artificial Intelligence in Smart Agriculture: A Review." In Lecture Notes on Data Engineering and Communications Technologies, Springer (2021). https://doi.org/10.1007/978-981-15-9793-0_44.

[7] Shah, S., Ghulam, M. A., Hareef, A., Muhammad, N., Gulzar, H., Mujahid, M., Rahim, B., Mahmoud, E., and Hosam, E. "Application of Drone Surveillance for Advance Agriculture Monitoring by Android Application Using Convolution Neural Network." Agronomy 13, no. 7 (2023). https://doi.org/10.3390/agronomy13071764.

[8] Velasco-Muñoz, J., Aznar-Sánchez, J., Batlles-delaFuente, A., and Fidelibus, M. "Sustainable Irrigation in Agriculture: An Analysis of Global Research." Water 11, no. 9 (2019). https://doi.org/10.3390/w11091758.

[9] Jeong, H., Kim, H., and Jang, T. "Irrigation Water Quality Standards for Indirect Wastewater Reuse in Agriculture: A Contribution toward Sustainable Wastewater Reuse in South Korea." Water 8, no. 4 (2016). https://doi.org/10.3390/w8040169.

[10] Seyedzadeh, A., Maroufpoor, S., Maroufpoor, E., Shiri, J., Bozorg-Haddad, O., and Gavazi, F. "Artificial Intelligence Approach to Estimate Discharge of Drip Tape Irrigation based on Temperature and Pressure." Agricultural Water Management 228 (2020). https://doi.org/10.1016/j.agwat.2019.105905.

[11] Sadiq, M., Singh, I., Ahmad, M., and Karunakaran, N. "The Internet of Things (IoT) for Sustainable Agriculture." In The Digital Agricultural Revolution, Wiley (2022). https://doi.org/10.1002/9781119823469.ch8.

[12] Talaviya, T., Shah, D., Patel, N., Yagnik, H., and Shah, M. "Implementation of Artificial Intelligence in Agriculture for Optimization of Irrigation and Application of Pesticides and Herbicides." Artificial Intelligence in Agriculture 4 (2020). https://doi.org/https://10.1016/j.aiia.2019.11.003.

[13] Pham, H., Awange, J., Kuhn, M., Nguyen, B., and Bui, L. "Enhancing Crop Yield Prediction Utilizing Machine Learning on Satellite-Based Vegetation Health Indices." Sensors 22, no. 3 (2022). https://doi.org/10.3390/s22030719.

[14] Perea, R., Poyato, E., Montesinos, P., and Díaz, J. "Prediction of Applied Irrigation Depths at Farm Level using Artificial Intelligence Techniques." Agricultural Water Management 206 (2018). https://doi.org/10.1016/j.agwat.2018.05.019.

[15] Kwaghtyo, D., and Eke, C. "Smart Farming Prediction Models for Precision Agriculture: A Complete Survey." Artificial Intelligence Review 56, no. 6 (2022). https://doi.org/https://10.1007/s10462-022-10266-6.

[16] Open Government Data Platform India. Accessible: https://data.gov.in. Accessed on March 15, 2023.

[17] Peterson, K. "The Acceptable R-square in Empirical Modelling for Social Science Research." MPRA (2023). Accessible: https://mpra.ub.uni-muenchen.de/115769/.

[18] McKenzie, J. "Mean Absolute Percentage Error and Bias in Economic Forecasting." Economics Letters 113, no. 3 (2011). https://doi.org/https://10.1016/j.econlet.2011.08.010.